TEACHING PROTOTYPE

Spatiotemporal Urban Activity Patterns

A hands-on prototype for students — Zurich, 2025

Based on: Barrena-Herrán, Modrego-Monforte & Grijalba (2025). *Revealing Spatiotemporal Urban Activity Patterns.* ISPRS Int. J. Geo-Inf. 14, 221.

Motivation

Why study the temporal rhythm of cities?

Where people concentrate during the day shapes almost everything in a city — transport demand, noise, retail turnover, public safety, air quality.

Classical urban planning focused on space. Time-geography (Hägerstrand, 1970) added the temporal dimension. Today, location-based data lets us measure both at scale.

24/7

Cities are now measured continuously

5

Stages in the reference methodology

k-shape

Clustering on normalised profile shape

The paper in one slide

Barrena-Herrán, Modrego-Monforte & Grijalba (2025)

Study area · Donostia-San Sebastián (ES).

Data · Google Popular Times for 1,378 POIs, one typical Friday.

Unit · Morphological grid (Voronoi cells around street intersections).

Method · Hourly weighted occupancy → k-shape → NNA + KDE.

Result · 5 distinct daily rhythm clusters with clear spatial footprints.

KEY INSIGHT

Urban form and daily rhythm are coupled — medieval cores, retail streets and residential neighbourhoods each have their own temporal fingerprint.

→ Our job today: reproduce this with open Zurich data.

Methodology

The five-stage pipeline

01

Collect POI + GPT data

02

Define the unit of analysis

03

Weighted hourly occupancy

04

Time-series clustering

05

Spatial analysis

This deck follows the same five stages — but uses an open, reproducible dataset that students can run end-to-end in an afternoon.

Our Zurich dataset (2025)

Open pedestrian & bicycle counts from Stadt Zürich

25	207k	15 min	12 mo
Active sensors	Hourly rows	Native resolution	Full year 2025

Source files

```
2025_verkehrszaehlungen_werte_fussgaenger_velo.csv
taz.view_eco_standorte.csv
```

Measured quantities

Directional bike & pedestrian counts (IN / OUT).

Why substitute?

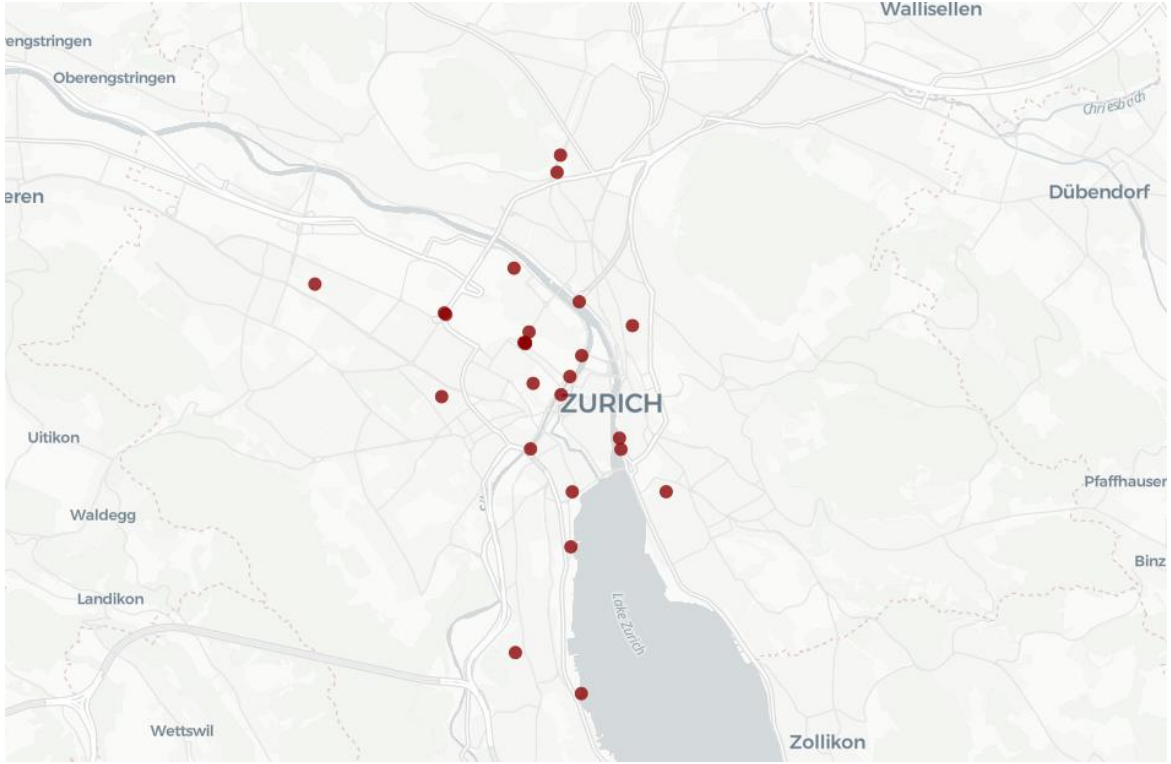
Google Popular Times requires scraping. Municipal sensor data is open, licence-clean, reproducible and exposes the same temporal structure we want to teach.

Trade-off

Fewer spatial units (25 vs. ~570) but a full year of data — ideal for teaching time-series clustering.

Counting stations

25 active pedestrian & bicycle sensors across the city

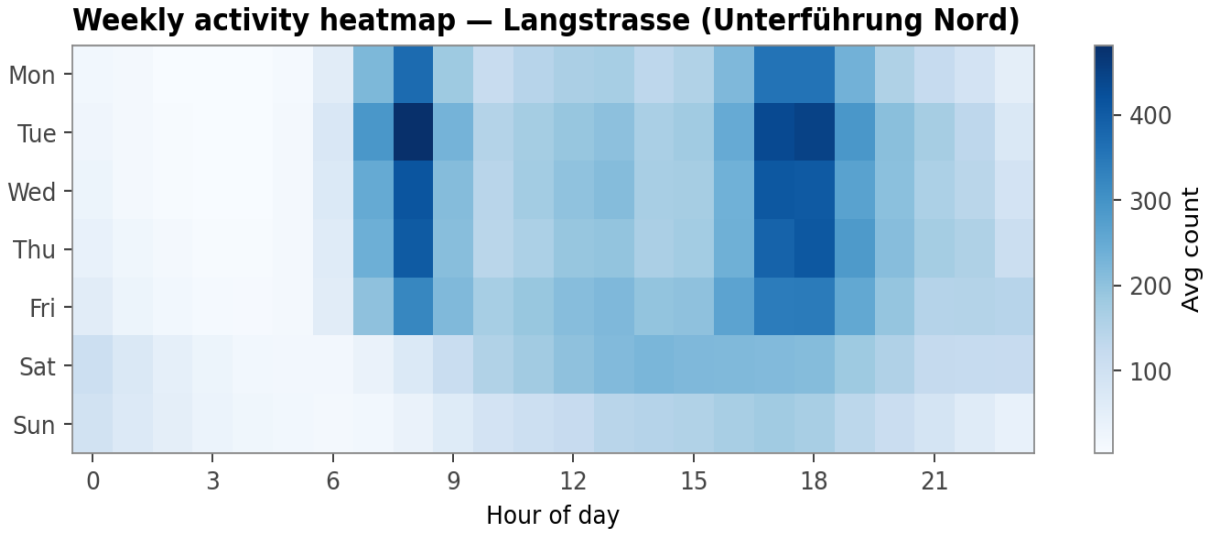


Concentration in the city centre (Limmat axis, Bahnhofstrasse area) and sparser coverage in residential districts.

Most sensors count bikes in both directions; a subset (~6) is pedestrian-only.

Exploratory analysis

Weekly activity heatmap — day of week × hour



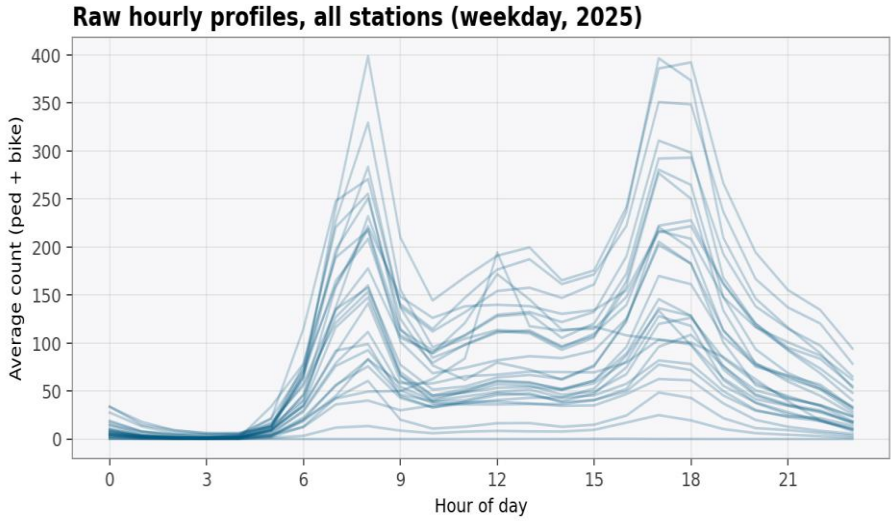
What to look for

- Weekday bimodal peaks → commuter site
- Saturday midday peak → shopping street
- Warm Friday evenings → leisure / nightlife

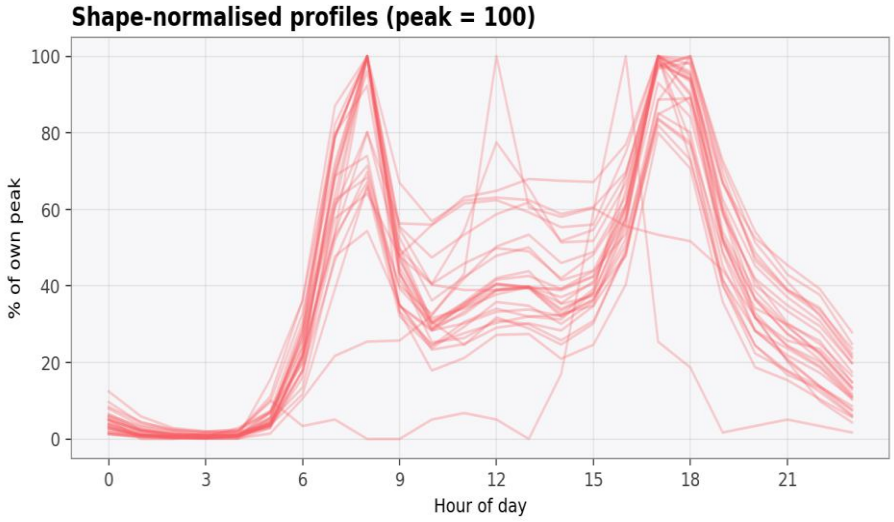
Before clustering, always eyeball the data. Weekly heatmaps let students spot commuter, retail and leisure patterns without any maths.

From counts to fingerprints

Stage 3 → 4 — why we normalise time series



Absolute magnitudes (problem)

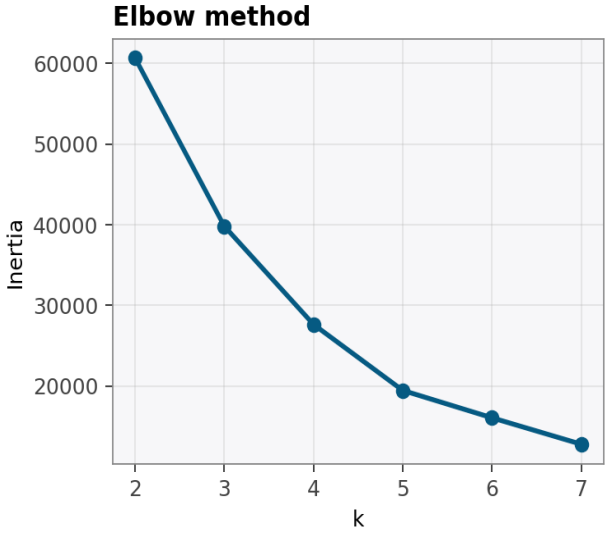


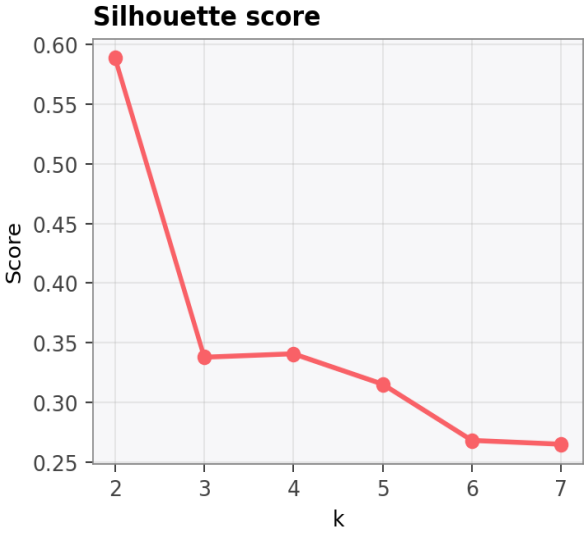
Divide by peak → shape only (solution)

A quiet residential sensor and a busy central sensor can have the exact same shape — two peaks at 8 & 18. After dividing by peak, they live on the same [0, 100] scale and the clustering compares what actually matters: the rhythm.

Model selection

Choosing k — elbow + silhouette





Elbow method
Pick the k where adding clusters stops paying off — here k ≈ 4.

Silhouette
Score in [-1, 1]. Higher means denser, better-separated clusters.

Both diagnostics agree that small k (3–4) already captures most of the structure in our 25-sensor dataset.

Clustering algorithm

k-means (today) vs. k-shape (paper)

k-means

- Minimises Euclidean distance to centroids
- Works on raw or normalised vectors
- Sensitive to time shifts (8:00 ≠ 9:00 peak)
- Built-in to sklearn — fast, simple, ideal for teaching

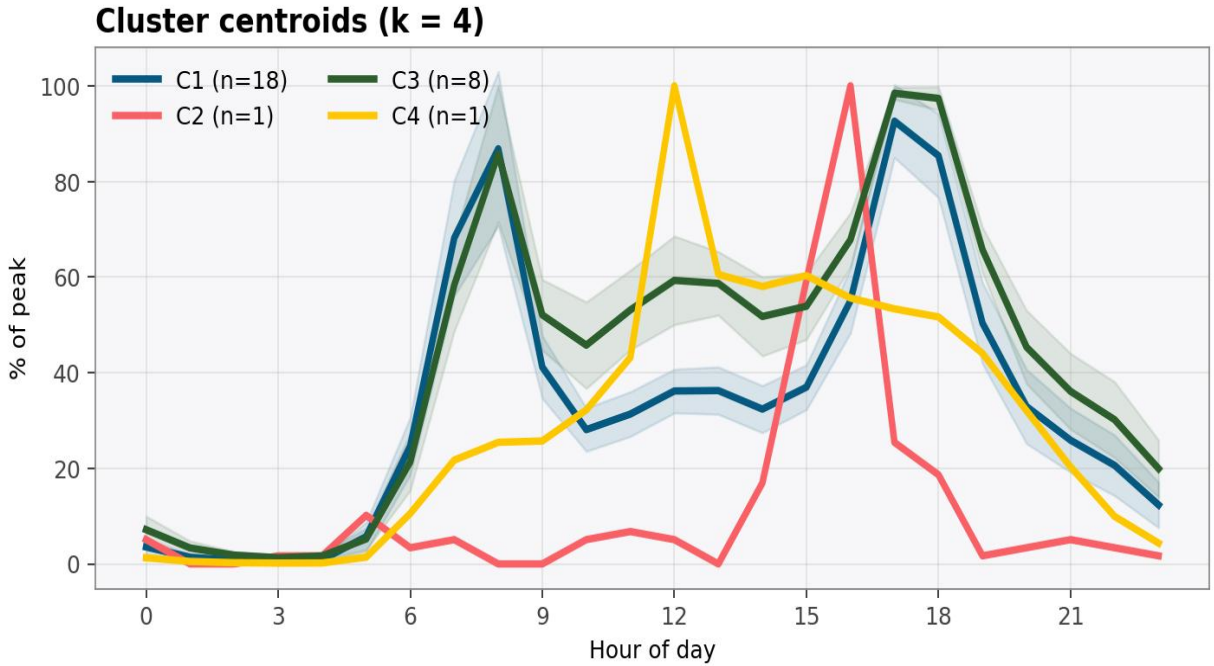
k-shape

- Shape-based distance (cross-correlation)
- Invariant to amplitude AND phase shift
- Better for time-series → chosen in the paper
- In tslearn — drop-in replacement in app.py

Teaching strategy: start with k-means (transparent), then let students swap it for k-shape and observe the difference.

Results — temporal clusters

Four distinct Zurich daily rhythms



C1 — Classic commuter

Twin peaks, 8h & 17–18h

C2 — Flat residential

Small, late-night outlier

C3 — Steady hotspot

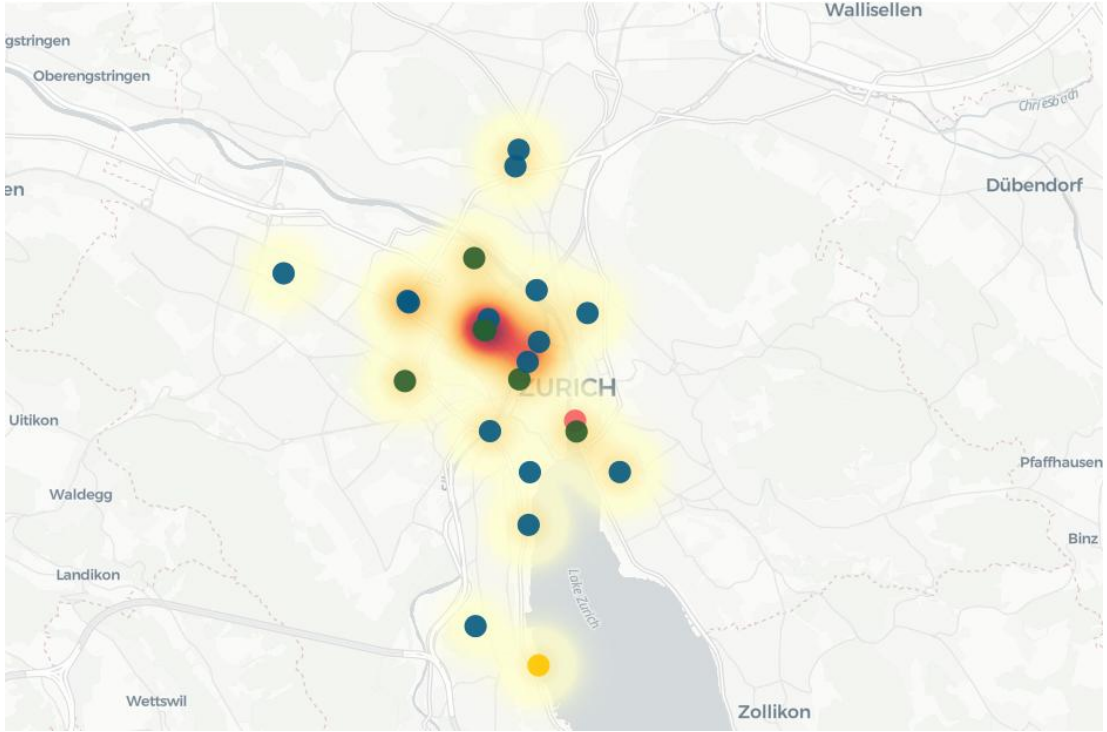
Strong all-day activity

C4 — Lunchtime leisure

Midday bump, no morning

Results — spatial distribution

Where does each rhythm live in Zurich?

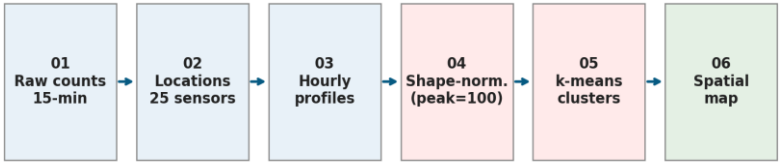


What to notice

- Commuter cluster dominates peripheral bike corridors.
- Hotspot cluster concentrates in the CBD and along the Limmat.
- Outlier stations stand out — often night-time or sparse sites.

Prototype

The accompanying Streamlit app



Six pages, one sidebar of controls:

1 · Introduction — paper summary & pipeline
2 · Dataset exploration — interactive map + heatmap
3 · Temporal profiles — raw vs. normalised
4 · Clustering — elbow, silhouette, centroids
5 · Spatial analysis — map coloured by cluster
6 · Exercises — Level 1–4 student tasks

Sidebar: modality, day type, month range, k.

R U N

```
streamlit run app.py
```

Requires: pandas, scikit-learn, streamlit, altair, pydeck

All data (CSV) bundled in /data — no download step.

PART 2

Student exercises

Four levels — from sidebar tweaking to method swapping.

Level 1 — Reproduce

Warm up: exploring the sidebar

01

Modality sensitivity

Switch between pedestrian-only and bike-only. Which stations cluster differently?

02

Weekday vs. weekend

Some rhythms invert — identify a 'commuter on weekdays, leisure on weekends' sensor.

03

Seasonality check

Summer (6–8) vs. winter (12–2): which clusters are robust, which change?

Level 2 — Explain

From clusters to narratives

04 · Name each cluster centroid (k = 4) in one sentence — 'commuter', 'evening peak', 'flat', etc.

05 · Pick two stations from your map and justify their cluster assignment given their land-use context (check on OpenStreetMap).

06 · Write a 100-word planning takeaway: which neighbourhoods might benefit from intervention based on what the rhythms reveal?

DELIVERABLE

A 1-page 'Temporal fingerprints of my neighbourhood' briefing with the cluster map, 4 centroids and 3 sentences of interpretation per cluster.

Submit as PDF.

Level 3 — Extend

Swap the algorithm, add dimensions

Hierarchical

Replace KMeans with AgglomerativeClustering. Plot the dendrogram.

DTW / k-shape

Install tslearn, swap in KShape — compare cluster memberships.

PCA embedding

Project 24-dim profiles to 2-D, colour by cluster, check separability.

Stability

Run 20 seeds; measure how often two sensors end up together.

Level 4 — Connect

Bridge back to the paper

10 · Build a toy example: two sine waves offset by 1 hour. Show that k-means assigns them to different clusters, but k-shape does not.

11 · Read Section 3.2 of the paper. Map their five clusters to the ones you found in Zurich — which correspond and which have no Zurich equivalent?

12 · The paper weights cells by capacity (people/m²). Add a weight to our k-means based on total counts — does the clustering change?

Summary

Five things to take home

1. **Cities have rhythms.**
   Normalised hourly profiles reveal structure that raw counts hide.

2. **Shape beats scale.**
   Normalising to peak lets a quiet alley share a cluster with a busy square if their rhythms match.

3. **Pick k with diagnostics.**
   Elbow + silhouette give you a defensible choice — not a guess.

4. **Algorithm matters.**
   k-means is a starting point; k-shape and DTW fit time-series better.

5. **Pipelines transfer.**
   The paper's methodology works on any spatiotemporal activity dataset.

THANK YOU

Questions & discussion

REFERENCES & DATA

Barrena-Herrán, M., Modrego-Monforte, I. & Grijalba, O. (2025). *Revealing Spatiotemporal Urban Activity Patterns: A Machine Learning Study Using Google Popular Times.* ISPRS Int. J. Geo-Inf., 14, 221.
Stadt Zürich Open Data · Verkehrszählungen Fussgänger & Velo 2025.